

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ K .Deeksha Sree(192424097)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

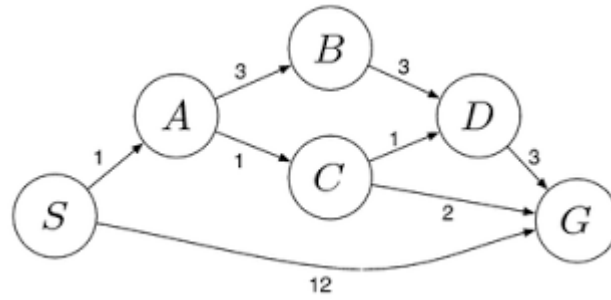
Signature of the student: K .Deeksha Sree

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. Given:

- Jug A = 4 gallons
 - Jug B = 3 gallons
 - Unlimited water supply
 - Goal: Obtain exactly 2 gallons in the 4-gallon jug
- Use the **State Space Search** technique.

Represent each state as:

$$(A, B)$$

where:

- A = amount of water in the 4-gallon jug
- B = amount of water in the 3-gallon jug

Initial State:

$$(0, 0)$$

Goal State:

$$(2, x)$$

where x can be 0–3.

Step	Action	State (4-gal,3-gal)
Initial	Empty jugs	(0,0)
1	Fill 3-gallon jug	(0,3)
2	Pour into 4-gallon jug	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4-gallon jug	(2,0)

Goal achieved:

$$(2, 0)$$

Exactly **2 gallons** are in the **4-gallon jug**.

2. The Mars Rover receives:

- Camera images
- Rock samples
- Soil composition
- Temperature
- Atmospheric pressure
- Obstacle detection
- GPS/location
- Wheel status
- Battery level

Property	Type
Observable	Partially Observable
Agents	Single Agent

Property	Type
Environment	Dynamic
Nature	Continuous
Knowledge	Unknown
Decision	Sequential

- ☐ Move Forward
- ☐ Move Backward
- ☐ Turn Left
- ☐ Turn Right
- ☐ Collect Rock Sample
- ☐ Drill Surface
- ☐ Capture Images
- ☐ Analyze Soil
- ☐ Send Data to Earth
- ☐ Recharge Solar Panels

Performance is evaluated using:

- Number of samples collected
- Accuracy of analysis
- Distance covered
- Battery efficiency
- Number of obstacles avoided
- Successful communication with Earth
- Mission completion rate

The rover must:

- Reach scientific goals
- Avoid obstacles
- Save battery
- Maximize scientific discoveries

Utility-based reasoning helps choose the best action among many alternatives.

3. Place 8 queens on an 8×8 chessboard so that:

- No two queens share the same row.
- No two queens share the same column.
- No two queens share the same diagonal.

Initial State

Empty chessboard.

State

A partially filled chessboard with queens placed safely.

Actions

Place one queen in a safe position of the next column.

Goal Test

Eight queens placed without attacking each other.

Path Cost

Each move costs 1.

Total cost = 8.

Use:

- Backtracking Search
- Depth First Search (DFS)

Search Space

Each column has 8 possible positions.

Maximum search space:

$$8^8 = 16,777,216$$

Backtracking greatly reduces this by pruning invalid states.

Backtracking efficiently explores only valid queen placements until a complete solution is found.

4. Customer selects:

- Pickup
- Destination
- Cab type

System books the best available cab.

Reason:

The goal is to transport the customer safely from source to destination

START

Input Pickup Location

Input Destination

Display Cab Types

(Mini, Micro, Sedan, Prime, Shared)

User selects Cab

Check Availability

IF cab available THEN

 Calculate Fare

 Assign Driver

 Confirm Booking

ELSE

 Display "Cab Not Available"

END IF

END

The Goal-Based Agent searches for an available cab that satisfies the customer's destination and preferred cab type.

5. Find the least-cost path from warehouse S to warehouse G.

Use Uniform Cost Search (UCS).

Reason:

UCS always expands the node with the lowest cumulative path cost.

- 1. Insert start node S into Priority Queue.**
- 2. Remove node having smallest cost.**
- 3. If goal reached, stop.**
- 4. Expand neighbours.**
- 5. Add neighbour with cumulative cost.**
- 6. Repeat until goal found.**

UniformCostSearch(Graph, Start, Goal)

PriorityQueue $\leftarrow \{(Start,0)\}$

Visited $\leftarrow$ Empty

while Queue not empty

Remove node with lowest cost

if node == Goal

return path

for each neighbour

Add neighbour with total cost

return Failure

Node Cost

S 0

A 3

B 5

C 7

G 9

Least-cost path:

S → A → C → G

Total Cost:

$$3 + 4 + 2 = 9$$

Uniform Cost Search guarantees the optimal path because it always expands the lowest cumulative cost node first.